JAVA PROGRAMMING

Complete MCQ Bank — Master Edition

288 Questions | All Topics | Answers Highlighted

Question Distribution: Exceptions: 13 | OOP: 15 | Keywords: 16 | Access Modifiers: 5 | Strings: 11 | Interfaces: 7 | Data Types: 13 | Arrays & Collections: 5 | Wrapper Classes: 7 | Java Basics: 19 | Constructors: 6 | Advanced: 100 | Lambda & Functional Interfaces: 9 | Streams API: 18 | Generics & Collections Framework: 4 | Advanced Lambda & Streams: 40

▶ EXCEPTIONS (13 Questions)

Q1 Which exception occurs when converting an invalid string to a number?

a. ClassCastException
b. ArithmeticException
c. NumberFormatException ✓
d. IOException

Q2 Which block is used to place code that might generate an exception?

a. try ✓
b. catch
c. throw
d. finally

Q3 Which block is used to handle exceptions?

a. catch ✓
b. finally
c. try
d. throw

Q4 Which class is the parent of all exception classes?

a. RuntimeException
b. Exception
c. Error
d. Throwable ✓

Q5 Which exception occurs when accessing a null object?

a. IOException
b. ClassNotFoundException
c. ArithmeticException
d. NullPointerException ✓

Q6 **Which exception occurs when array index is out of range?**

a. OutOfBoundsError
b. IndexException
c. ArrayIndexOutOfBoundsException ✓
d. ArrayException

Q7 **Which exception occurs when dividing a number by zero?**

a. NullPointerException
b. IOException
c. ArithmeticException ✓
d. NumberFormatException

Q8 **Which keyword is used to explicitly throw an exception?**

a. try
b. throws
c. throw ✓
d. catch

Q9 **Which keyword declares exceptions that a method may throw?**

a. throws ✓
b. catch
c. throw
d. finally

Q10 **Which keyword is used before a method to indicate it may throw an exception?**

a. throw
b. catch
c. throws ✓
d. try

Q11 **What is the purpose of the 'finally' block?**

a. To handle exceptions
b. To throw exceptions
c. To execute code regardless of exception ✓
d. To catch specific exceptions

Q12 **Which of the following is a checked exception?**

a. NullPointerException
b. ArrayIndexOutOfBoundsException

c. **IOException ✓**
d. ClassCastException

Q13 **Which of the following is an unchecked exception?**
a. IOException
b. FileNotFoundException
c. SQLException
d. **NullPointerException ✓**

▶ OOP (15 Questions)

Q14 **Which one of the following is NOT a OOPs Pillar?**
a. Polymorphism
b. Abstraction
c. Inheritance
d. **Exception ✓**

Q15 **Which concept is NOT a major OOP concept?**
a. **Compilation ✓**
b. Encapsulation
c. Inheritance
d. Polymorphism

Q16 **What is encapsulation?**
a. **Hiding implementation details and exposing only necessary interfaces ✓**
b. Inheriting from parent
c. Creating multiple objects
d. Overloading methods

Q17 **What is abstraction in OOP?**
a. Exposing all class details
b. **Hiding complexity and showing only essential features ✓**
c. Copying class data
d. Merging two classes

Q18 **Which of the following variable violates the definition of encapsulation?**
a. Array variables
b. **Public variables ✓**
c. Local variables

d. Global variables

Q19 **If a class encapsulates data and methods, what is it preventing?**

a. Code reusability
b. Direct manipulation of its data ✓
c. Implementation of interfaces
d. Use of inheritance

Q20 **What is polymorphism in Java?**

a. A class having multiple constructors
b. One interface, multiple implementations ✓
c. A class extending multiple classes
d. Variables with multiple types

Q21 **Which type of polymorphism is achieved through method overloading?**

a. Late binding
b. Compile-time polymorphism ✓
c. Dynamic polymorphism
d. Runtime polymorphism

Q22 **What is runtime polymorphism also known as?**

a. Static dispatch
b. Compile-time polymorphism
c. Dynamic dispatch via method overriding ✓
d. Method overloading

Q23 **Early binding is also known as?**

a. Dynamic binding
b. Late binding
c. Run time binding
d. Static binding ✓

Q24 **What is method overriding in Java?**

a. Defining method with same name and different parameters
b. Defining method in child with same name and signature as parent ✓
c. Creating two constructors
d. Calling superclass method

Q25 **What is method overloading in Java?**

a. Defining same method name with different parameters in same class ✓

b. Redefining a method from parent class
c. Creating a method in interface
d. Using abstract methods

Q26 **In which type of inheritance does one class act as a superclass for more than one sub-class?**

a. Multilevel inheritance
b. Multiple inheritance
c. Hierarchical inheritance ✓
d. Hybrid inheritance

Q27 **Which supports multiple inheritance in Java?**

a. Abstract class
b. Class
c. Interface ✓
d. Enum

Q28 **What is the diamond problem in Java?**

a. Memory leak
b. Ambiguity from multiple inheritance ✓
c. Infinite loop
d. Null pointer issue

▶ KEYWORDS (16 Questions)

Q29 **Which keyword is used to prevent a method from being overridden?**

a. private
b. abstract
c. static
d. final ✓

Q30 **What is the purpose of the 'final' keyword in Java?**

a. All of mentioned ✓
b. To prevent modification of variable's value
c. To prevent overriding of a method
d. To prevent inheritance of a class

Q31 **Which keyword prevents a class from being inherited?**

a. private
b. static

c. **final ✓**
d. abstract

Q32 **What is the purpose of the 'super' keyword in Java?**
a. To hide data within a class
b. To create multiple instances
c. To refer to the current object
d. **To invoke the superclass constructor or methods ✓**

Q33 **Which keyword refers to the current object?**
a. **this ✓**
b. current
c. super
d. self

Q34 **Which keyword refers to the parent class object?**
a. **super ✓**
b. base
c. parent
d. this

Q35 **Which keyword is used to create an object?**
a. make
b. object
c. **new ✓**
d. create

Q36 **Which keyword is used to define a class?**
a. define
b. object
c. **class ✓**
d. struct

Q37 **Which keyword is used to define an abstract class?**
a. **abstract ✓**
b. interface
c. virtual
d. final

Q38 **Which keyword is used to implement an interface?**

a. extends
b. implements ✓
c. interface
d. inherit

Q39 Which keyword is used to inherit a class?
a. implement
b. extends ✓
c. import
d. inherits

Q40 Which keyword is used to define a package in Java?
a. package ✓
b. import
c. class
d. define

Q41 Which keyword is used to access classes from another package?
a. include
b. package
c. import ✓
d. using

Q42 Which keyword is used to create a subclass exception?
a. throw
b. implement
c. inherit
d. extends ✓

Q43 What does the 'static' keyword mean in Java?
a. Member belongs to an instance
b. Member belongs to the class ✓
c. Member is constant
d. Member is private

Q44 Which keyword is used to define a constant in Java?
a. const
b. constant
c. static final ✓
d. define

▶ ACCESS MODIFIERS (5 Questions)

Q45 **Which access modifier allows access within the same class only?**

- **a. private ✓**
- b. default
- c. protected
- d. public

Q46 **Which keyword makes a variable accessible to all classes?**

- a. private
- b. default
- c. protected
- **d. public ✓**

Q47 **What is the 'protected' access modifier?**

- a. Accessible only within same class
- b. Accessible only within same package
- **c. Accessible within same package and subclasses ✓**
- d. Accessible everywhere

Q48 **Which member of the superclass is never accessible to the subclass?**

- a. Public member
- b. Protected member
- c. All of the mentioned
- **d. Private member ✓**

Q49 **Which of the following is NOT a Java access modifier?**

- a. public
- b. private
- **c. friendly ✓**
- d. protected

▶ STRINGS (11 Questions)

Q50 **Which class provides thread-safe mutable strings?**

- a. CharBuffer
- **b. StringBuffer ✓**
- c. String

d. StringBuilder

Q51 **Which class is used to create mutable strings?**

a. Character

b. StringBuilder ✓

c. StringConstant

d. String

Q52 **Which class represents immutable strings in Java?**

a. CharSequence

b. StringBuilder

c. String ✓

d. StringBuffer

Q53 **Which method is used to convert a String to uppercase?**

a. toUpperCase() ✓

b. toUpper()

c. upperCase()

d. convertUpper()

Q54 **Which method is used to find the length of a String?**

a. length() ✓

b. getLength()

c. size()

d. count()

Q55 **What is the output: String str1='cdac'; String str2=new String('cdac'); System.out.println(str1==str2);**

a. Equal

b. Not Equal

c. false ✓

d. true

Q56 **In which memory is a String stored when created using 'new' operator?**

a. Stack

b. Random storage space

c. String memory

d. Heap memory ✓

Q57 **Which String method removes leading and trailing whitespace?**

a. trim()
b. strip()
c. clean()
d. Both trim() and strip() ✓

Q58 Which String method checks if a string starts with a prefix?
a. beginsWith()
b. startsWith() ✓
c. hasPrefix()
d. prefix()

Q59 Which method is used to compare two strings ignoring case?
a. equals()
b. compareTo()
c. equalsIgnoreCase() ✓
d. compareIgnoreCase()

Q60 Which method returns a substring from a String?
a. split()
b. slice()
c. substring() ✓
d. sub()

▶ INTERFACES (7 Questions)

Q61 An interface with no fields or methods is known as a _____
a. Marker Interface ✓
b. Abstract Interface
c. Runnable Interface
d. CharSequence Interface

Q62 What is the purpose of a marker interface in Java?
a. To define methods implementing classes must implement
b. To indicate certain behavior or characteristics ✓
c. To mark a class as abstract
d. To mark a class as final

Q63 Which of the following statements is true about default methods in interfaces?
a. Default methods cannot be declared in an interface

b. Cannot be overridden by implementing classes
c. Must be implemented by implementing classes
d. Provide a default implementation for an interface method ✓

Q64 What is an abstract class?
a. A class with only static methods
b. A class that cannot be instantiated ✓
c. A class with only private methods
d. A class that cannot inherit

Q65 Can an abstract class have a constructor?
a. No, never
b. Yes, always ✓
c. Only if it has no abstract methods
d. Only if it is final

Q66 How many interfaces can a Java class implement?
a. Only one
b. Only two
c. Up to 5
d. Multiple ✓

Q67 What is a functional interface?
a. Interface with multiple abstract methods
b. Interface with no methods
c. Interface with exactly one abstract method ✓
d. Interface with only static methods

▶ DATA TYPES (13 Questions)

Q68 Which data type stores true or false values?
a. boolean ✓
b. int
c. char
d. float

Q69 Which is a non-primitive data type?
a. double
b. int

c. **String ✓**
d. float

Q70 **Which of the following is a primitive data type in Java?**
a. Scanner
b. Integer
c. String
d. **int ✓**

Q71 **Which primitive data type stores decimal numbers?**
a. boolean
b. char
c. **float ✓**
d. int

Q72 **Which is an example of widening conversion?**
a. **int to double ✓**
b. float to byte
c. double to int
d. long to short

Q73 **Which is an example of narrowing conversion?**
a. int to long
b. byte to int
c. **double to int ✓**
d. char to int

Q74 **Explicit type conversion is also known as:**
a. Automatic casting
b. Conversion error
c. Implicit casting
d. **Type casting ✓**

Q75 **Implicit type conversion is also called:**
a. Explicit casting
b. **Automatic casting ✓**
c. Narrowing
d. Manual casting

Q76 **What is the difference between static and instance variables?**

a. **Static belong to class; instance belong to object ✓**
b. No difference
c. Static are always final
d. Instance cannot be initialized

Q77 **Variables defined inside a class but outside methods are called:**
a. **Instance variables ✓**
b. Local variables
c. Parameters
d. Static blocks

Q78 **What is the size of an int in Java?**
a. 8 bits
b. 16 bits
c. **32 bits ✓**
d. 64 bits

Q79 **What is the default value of an int variable in Java?**
a. null
b. 1
c. -1
d. **0 ✓**

Q80 **What is the default value of a boolean variable in Java?**
a. true
b. 1
c. null
d. **false ✓**

▶ ARRAYS & COLLECTIONS (5 Questions)

Q81 **Which of the following is true about arrays in Java?**
a. Arrays do not store objects
b. Arrays are dynamically resized automatically
c. **Array size is fixed after creation ✓**
d. Arrays can store different data types

Q82 **Which collection does NOT allow duplicate elements?**
a. ArrayList

b. LinkedList
c. HashSet ✓
d. Vector

Q83 Which interface does ArrayList implement?
a. Set
b. Map
c. Queue
d. List ✓

Q84 Which Map implementation maintains insertion order?
a. HashMap
b. TreeMap
c. LinkedHashMap ✓
d. Hashtable

Q85 Which Map implementation sorts keys in natural order?
a. HashMap
b. LinkedHashMap
c. Hashtable
d. TreeMap ✓

▶ WRAPPER CLASSES (7 Questions)

Q86 Boxing means:
a. Converting primitive to wrapper object ✓
b. Removing object
c. Converting wrapper to primitive
d. Creating class

Q87 Unboxing means:
a. Method to Object
b. Object to Primitive ✓
c. Primitive to Object
d. Class to Object

Q88 Wrapper class for int is:
a. Number
b. Integer ✓

c. IntWrapper
d. Int

Q89 **Wrapper class for float is:**

a. Real
b. Float ✓
c. Decimal
d. Double

Q90 **Wrapper classes belong to which package?**

a. java.io
b. java.util
c. java.sql
d. java.lang ✓

Q91 **Which statement creates an Integer object from int?**

a. int obj = Integer(5);
b. Integer obj = 5.5;
c. Integer obj = "5";
d. Integer obj = new Integer(5); ✓

Q92 **What is autoboxing in Java?**

a. Automatic conversion of int to String
b. Automatic conversion of primitive to wrapper class ✓
c. Automatic conversion of wrapper to primitive
d. Automatic garbage collection

▶ JAVA BASICS (19 Questions)

Q93 **Java is best described as a:**

a. Operating System
b. Programming Language, Platform, and Technology ✓
c. Database System
d. Programming Language only

Q94 **Java was initially called:**

a. Pine
b. Coffee
c. Green

d. **Oak ✓**

Q95 **Which company originally developed Java?**

a. Oracle
b. IBM
c. Microsoft
d. **Sun Microsystems ✓**

Q96 **Which company currently maintains Java?**

a. Microsoft
b. **Oracle ✓**
c. Google
d. IBM

Q97 **What does JVM stand for?**

a. Java Variable Method
b. **Java Virtual Machine ✓**
c. Java Verified Module
d. Java Version Manager

Q98 **What does JDK stand for?**

a. **Java Development Kit ✓**
b. Java Default Kernel
c. Java Data Key
d. Java Dynamic Kit

Q99 **What does JRE stand for?**

a. Java Runtime Execution
b. **Java Runtime Environment ✓**
c. Java Real Engine
d. Java Recursive Engine

Q100 **Which component is used to compile Java programs?**

a. **JDK ✓**
b. JVM
c. JIT
d. JRE

Q101 **Which component is responsible for converting bytecode into machine specific code?**

a. JRE

b. JVM

c. JIT ✓

d. JDK

Q102 Which file extension represents compiled Java bytecode?

a. .java

b. .class ✓

c. .exe

d. .jar

Q103 Which command compiles a Java program?

a. javac ✓

b. run

c. compile

d. java

Q104 Which command runs a compiled Java program?

a. javac

b. execute

c. java ✓

d. compile

Q105 Correct signature of the main method is:

a. public static void main(String args[]) ✓

b. void main()

c. public void main()

d. static void main()

Q106 Which method is the entry point of a Java program?

a. start()

b. run()

c. main() ✓

d. init()

Q107 Every Java program must contain:

a. Class ✓

b. Package

c. Interface

d. Constructor

Q108 **Which package is automatically imported in every Java program?**

a. java.io
b. java.util
c. java.lang ✓
d. java.net

Q109 **Out of these packages, which one contains the abstract keyword?**

a. java.system
b. java.util
c. java.io
d. java.lang ✓

Q110 **Which stream is used for output?**

a. System.out ✓
b. System.read
c. System.in
d. System.write

Q111 **Which stream is used for standard input in Java?**

a. System.err
b. InputStream
c. System.in ✓
d. System.out

▶ CONSTRUCTORS (6 Questions)

Q112 **What is a constructor in Java?**

a. A method that returns a value
b. A special method called when an object is created ✓
c. A static method
d. An abstract method

Q113 **When an object is created, an initialization is automatically done by?**

a. Destructor
b. Friend
c. Constructor ✓
d. Member

Q114 **What is a default constructor?**

a. A constructor with parameters
b. A constructor provided by Java when no constructor is defined ✓
c. A static constructor
d. A final constructor

Q115 What is the parent class of all classes in Java?
a. Class
b. Super
c. Base
d. Object ✓

Q116 What is a singleton class?
a. A class with only one method
b. A class that allows only one instance ✓
c. A class with no constructors
d. A final class

Q117 Which method is used to get the class of an object at runtime?
a. getClass() ✓
b. classOf()
c. instanceof()
d. getType()

▶ ADVANCED (100 Questions)

Q118 What is the output of: System.out.println(1 + 2 + '3'); in Java?
a. 123
b. 51 ✓
c. 6
d. Error

Q119 What is the output of: System.out.println('A' + 1); in Java?
a. A1
b. 66 ✓
c. B
d. Error

Q120 Which of the following correctly declares a 2D array in Java?
a. int arr[][];

b. int[][] arr;
c. int arr = new int[3][3];
d. Both a and b ✓

Q121 What will happen if the 'catch' block is followed by a 'finally' block and the catch block throws an exception?

a. finally block is skipped
b. finally block still executes ✓
c. Program terminates immediately
d. Compile error occurs

Q122 Can a 'try' block exist without a 'catch' block in Java?

a. No, catch is mandatory
b. Yes, if there is a finally block ✓
c. Yes, always
d. Only in abstract classes

Q123 Which of the following will cause a compile-time error?

a. Catching Exception before RuntimeException ✓
b. Catching RuntimeException before Exception
c. Catching IOException after Exception
d. Catching both IOException and SQLException

Q124 What is a 'chained exception' in Java?

a. Multiple catch blocks
b. An exception that holds reference to another exception as its cause ✓
c. Two try blocks nested
d. Exception thrown in finally

Q125 Which method is used to get the cause of an exception?

a. getMessage()
b. getCause() ✓
c. getClass()
d. getStack()

Q126 What is the output of: int[] a = {1,2,3}; System.out.println(a[-1]);?

a. 0
b. null
c. ArrayIndexOutOfBoundsException ✓
d. Compile error

Q127 **A method in a subclass has the same name and parameters as a static method in the parent class. This is called?**

a. Method overriding
b. Method overloading
c. Method hiding ✓
d. Method shadowing

Q128 **Can a static method be overridden in Java?**

a. Yes, always
b. No, static methods can only be hidden, not overridden ✓
c. Only if they have same signature
d. Only in abstract classes

Q129 **Which of the following is true about constructor chaining?**

a. It uses the 'super' or 'this' keyword to call another constructor ✓
b. It creates infinite loops
c. It is done using 'new' keyword
d. Constructors cannot call other constructors

Q130 **What is the output of this code? class A { static int x=5; } class B extends A { static int x=10; } A obj = new B(); System.out.println(obj.x);**

a. 10
b. 5 ✓
c. 0
d. Compile error

Q131 **What does 'covariant return type' mean in Java?**

a. An overriding method can return a subtype of the return type declared in the parent ✓
b. Return type must exactly match the parent
c. Only primitive types can be covariant
d. Static methods support covariant return

Q132 **If a class implements two interfaces with the same default method, what must the class do?**

a. It will inherit from the first interface
b. It will inherit from the second interface
c. It must override the method to resolve ambiguity ✓
d. It causes a compile error automatically

Q133 **Which keyword is used in Java to handle multiple exceptions in a single catch block?**

a. ||

b. &&

c. | ✓

d. multi

Q134 What is the difference between Error and Exception in Java?

a. No difference

b. Errors are unrecoverable system-level problems; Exceptions can be handled ✓

c. Exceptions are always checked; Errors are unchecked

d. Errors extend Exception

Q135 Which of these exceptions is thrown when the JVM runs out of heap memory?

a. StackOverflowError

b. OutOfMemoryError ✓

c. HeapException

d. MemoryException

Q136 What is the output? String s = null; System.out.println(s + ' world');

a. Compile error

b. NullPointerException

c. null world ✓

d. world

Q137 What is the purpose of the 'transient' keyword?

a. To prevent garbage collection

b. To skip serialization of a field ✓

c. To make a field thread-safe

d. To define a constant

Q138 What is the purpose of the 'volatile' keyword?

a. Makes a variable immutable

b. Ensures visibility of variable changes across threads ✓

c. Prevents compilation

d. Defines a static variable

Q139 What is the purpose of the 'synchronized' keyword?

a. Increases performance

b. Allows only one thread at a time in a block ✓

c. Sorts collections

d. Implements interfaces

Q140 **Which design pattern ensures a class has only one instance?**

a. Factory
b. Observer
c. Singleton ✓
d. Decorator

Q141 **What happens when you call Thread.sleep(1000) inside a synchronized block?**

a. The lock is released
b. The lock is NOT released; the thread sleeps while holding it ✓
c. The thread terminates
d. A deadlock always occurs

Q142 **What is a deadlock in Java?**

a. A program that runs too fast
b. Two or more threads permanently blocked, each waiting for a lock held by the other ✓
c. An exception during file I/O
d. A garbage collection pause

Q143 **Which method is used to suggest garbage collection?**

a. System.collect()
b. System.gc() ✓
c. Runtime.clean()
d. Garbage.run()

Q144 **What is serialization in Java?**

a. Converting an object to a byte stream ✓
b. Converting byte stream to object
c. Sorting an object
d. Encrypting an object

Q145 **Which interface must a class implement to be serializable?**

a. Runnable
b. Serializable ✓
c. Cloneable
d. Comparable

Q146 **What is the purpose of serialVersionUID in Java?**

a. Helps JVM identify the version of the class during deserialization ✓
b. Acts as a primary key
c. Prevents serialization
d. Encrypts the object

Q147 **What does the instanceof operator do?**

a. Creates a new instance
b. Checks if an object is an instance of a class ✓
c. Destroys an instance
d. Counts instances

Q148 **What is the difference between == and .equals() in Java?**

a. No difference
b. == compares references; .equals() compares content ✓
c. == compares content; .equals() compares references
d. Both compare content

Q149 **What is the output of: Integer a=127; Integer b=127; System.out.println(a==b);?**

a. false
b. true ✓
c. Compile error
d. NullPointerException

Q150 **What is the output of: Integer a=128; Integer b=128; System.out.println(a==b);?**

a. true
b. false ✓
c. Compile error
d. NullPointerException

Q151 **What is Integer caching in Java?**

a. JVM caches Integer objects for values -128 to 127 and reuses them ✓
b. JVM caches all Integer objects
c. Integer values are stored in ROM
d. Caching improves String concatenation

Q152 **A lambda expression in Java is essentially an instance of a _____.**

a. Abstract class
b. Functional interface ✓
c. Generic class
d. Anonymous class only

Q153 **Which Java version introduced lambda expressions?**

a. Java 6
b. Java 7

c. **Java 8 ✓**
d. Java 9

Q154 **What does Stream.filter() return?**
a. A boolean
b. **A new filtered Stream ✓**
c. A List
d. void

Q155 **Which of the following is a terminal operation in the Stream API?**
a. filter()
b. map()
c. **collect() ✓**
d. sorted()

Q156 **What is method reference syntax in Java?**
a. Class->method
b. Class.method
c. **Class::method ✓**
d. Class@method

Q157 **What does Optional.orElse() do?**
a. Throws exception if empty
b. **Returns value if present, otherwise returns the given default ✓**
c. Always returns null
d. Converts Optional to Stream

Q158 **Which of the following correctly creates an immutable list in Java 9+?**
a. new ArrayList()
b. Arrays.asList()
c. **List.of() ✓**
d. Collections.unmodifiableList()

Q159 **What is the 'var' keyword introduced in Java 10?**
a. A dynamic variable type
b. **Local variable type inference — compiler infers the type ✓**
c. A global variable keyword
d. Replaces 'final'

Q160 **Which annotation indicates a method overrides a parent class method?**

a. @Super
b. @Overrides
c. @Override ✓
d. @Inherit

Q161 What is the @FunctionalInterface annotation used for?
a. Makes a class functional
b. Ensures interface has exactly one abstract method ✓
c. Creates lambda automatically
d. Makes interface final

Q162 What is the output of: for(int i=0;i<3;i++) { if(i==1) continue; System.out.print(i+' '); }?
a. 0 1 2
b. 0 2 ✓
c. 1 2
d. 0 1

Q163 What is the output of: int x=5; System.out.println(x++ + ++x);?
a. 10
b. 11
c. 12 ✓
d. 13

Q164 What is the output of: System.out.println(10 % 3);?
a. 3
b. 1 ✓
c. 0
d. 3.33

Q165 Which of the following correctly declares a generic class?
a. class Box<T> ✓
b. class Box[T]
c. class Box(T)
d. class Box{T}

Q166 What is the wildcard '?' used for in Java generics?
a. Unknown method
b. Unknown type parameter ✓
c. Unknown variable
d. Optional parameter

Q167	What is an inner class in Java?
a.	A class inside a method
b.	**A class defined inside another class ✓**
c.	A class with private constructor
d.	A class with no methods

Q168	What is a static nested class?
a.	A class declared inside a method
b.	**A class that can only access static members of the outer class ✓**
c.	An interface inside a class
d.	An abstract nested class

Q169	Which exception is thrown when calling a method on a null reference?
a.	IllegalArgumentException
b.	ClassCastException
c.	**NullPointerException ✓**
d.	IndexOutOfBoundsException

Q170	What is the difference between throw and throws in Java?
a.	No difference
b.	**'throw' is used to explicitly throw an exception; 'throws' declares exceptions a method may propagate ✓**
c.	'throws' explicitly throws; 'throw' declares
d.	They are interchangeable

Q171	What is a try-with-resources statement?
a.	A try block that creates resources
b.	**A try block that automatically closes AutoCloseable resources ✓**
c.	A try block with multiple catches
d.	A try block without finally

Q172	What interface must a resource implement to be used in try-with-resources?
a.	Closeable
b.	Serializable
c.	**AutoCloseable ✓**
d.	Runnable

Q173	What is the output of: System.out.println(0.1 + 0.2 == 0.3); in Java?
a.	true
b.	**false ✓**

c. Compile error
d. 0.3

Q174 **Which class should you use for precise decimal calculations (e.g., currency)?**

a. double
b. float
c. BigDecimal ✓
d. Math

Q175 **What is the output of: System.out.println(Integer.MAX_VALUE + 1);?**

a. 2147483648
b. Integer overflow: -2147483648 ✓
c. Compile error
d. Infinity

Q176 **Which of the following is NOT a valid way to iterate over a List in Java?**

a. for-each loop
b. Iterator
c. for loop with index
d. while loop using get(-1) ✓

Q177 **What does Collections.sort() use internally by default?**

a. Bubble sort
b. Merge sort (TimSort) ✓
c. Quick sort
d. Insertion sort

Q178 **What is the time complexity of HashMap.get() in the average case?**

a. O(n)
b. O(log n)
c. O(1) ✓
d. $O(n^2)$

Q179 **Which interface provides natural ordering in Java?**

a. Comparator
b. Iterable
c. Comparable ✓
d. Cloneable

Q180 **What is the difference between Comparable and Comparator?**

a. No difference

b. Comparable is for natural ordering (compareTo); Comparator is for custom ordering (compare) ✓

c. Comparator is in java.lang; Comparable in java.util

d. Comparable is used only for primitives

Q181 What is the output of: List<Integer> list = Arrays.asList(1,2,3); list.add(4);?

a. [1,2,3,4]

b. [4,1,2,3]

c. UnsupportedOperationException ✓

d. Compile error

Q182 What is a ConcurrentModificationException?

a. Thrown when a collection is modified while iterating over it ✓

b. Thrown when two threads read the same collection

c. Thrown when a null is added

d. Thrown when sorting fails

Q183 Which collection is thread-safe by default?

a. ArrayList

b. HashMap

c. LinkedList

d. Vector ✓

Q184 What does the 'final' keyword do when applied to a reference variable?

a. Makes the object immutable

b. Prevents the reference from pointing to a different object, but object state can still change ✓

c. Prevents any method calls on the object

d. Makes the variable static

Q185 Which feature of Java makes it platform-independent?

a. Object-oriented nature

b. Bytecode execution by JVM ✓

c. Use of pointers

d. Automatic garbage collection

Q186 What is the purpose of the 'assert' statement in Java?

a. To throw exceptions

b. To test assumptions during development ✓

c. To define constants

d. To override methods

Q187 **How do you enable assertions in Java at runtime?**

a. java -assert ClassName

b. java -ea ClassName ✓

c. java -assert:on ClassName

d. Assertions are enabled by default

Q188 **Which of the following is true about abstract methods?**

a. They have a method body

b. They must be declared in an abstract class or interface ✓

c. They can be private

d. They can be final

Q189 **What happens if you don't override all abstract methods in a concrete subclass?**

a. Runtime error

b. The class compiles fine

c. The class must also be declared abstract ✓

d. Partial implementation is allowed

Q190 **What is the output of: class A { int x=5; } class B extends A { int x=10; } B obj=new B(); System.out.println(obj.x);?**

a. 5

b. 10 ✓

c. 0

d. Compile error

Q191 **Which of the following correctly implements multiple inheritance behavior in Java?**

a. extends two classes

b. implements two interfaces ✓

c. extends and implements both

d. Using abstract classes only

Q192 **What is the output of: String s='hello'; s.toUpperCase(); System.out.println(s);?**

a. HELLO

b. hello ✓

c. Hello

d. Compile error

Q193 **Why does String s='hello'; s.toUpperCase(); NOT change s?**

a. toUpperCase() is broken

b. **Strings are immutable; toUpperCase() returns a new String ✓**
c. The method needs to be called on a new object
d. Uppercase is auto-reset

Q194 **What is intern() method in String?**
a. Converts to int
b. Converts to internal format
c. **Returns the canonical representation of the string from the String pool ✓**
d. Encrypts the string

Q195 **Which of the following creates a String in the String pool?**
a. String s = new String('hello');
b. **String s = 'hello'; ✓**
c. String s = String.create('hello');
d. Both a and b

Q196 **What is the result of: 'Hello'.concat(' World') == 'Hello World';?**
a. true
b. **false ✓**
c. Compile error
d. Depends on JVM

Q197 **What is the output of: System.out.println('abc'.charAt(5));?**
a. null
b. '\0'
c. **StringIndexOutOfBoundsException ✓**
d. Compile error

Q198 **What is the output of: StringBuilder sb=new StringBuilder('Hello'); sb.reverse(); System.out.println(sb);?**
a. Hello
b. **olleH ✓**
c. Error
d. null

Q199 **What does StringBuffer.append() return?**
a. void
b. String
c. A new StringBuffer
d. **The same StringBuffer object (for chaining) ✓**

Q200 **What is the difference between StringBuilder and StringBuffer?**

a. No difference
b. StringBuilder is faster but not thread-safe; StringBuffer is thread-safe but slower ✓
c. StringBuffer is faster
d. StringBuilder is immutable

Q201 **What is the output of: int x=10; int y=x>>>1; System.out.println(y);?**

a. 10
b. 5 ✓
c. 20
d. Error

Q202 **What does the >>> operator do in Java?**

a. Left shift
b. Right shift with sign extension
c. Unsigned right shift (fills with 0) ✓
d. Bitwise XOR

Q203 **What is the output of: System.out.println(true & false); in Java?**

a. true
b. false ✓
c. Compile error
d. 1

Q204 **What is the difference between & and && in Java?**

a. No difference
b. & always evaluates both operands; && short-circuits on false ✓
c. & is for integers only
d. && is bitwise

Q205 **What is the output of: int[] arr=new int[3]; System.out.println(arr[0]);?**

a. null
b. Undefined
c. 0 ✓
d. Compile error

Q206 **Which of the following is the correct way to pass an array to a method in Java?**

a. method(int arr[])
b. method(int[] arr)
c. method(arr[])

d. **Both a and b ✓**

Q207 **What is a vararg method in Java?**

a. A method with variable return type

b. **A method that accepts variable number of arguments ✓**

c. A method with no arguments

d. A virtual abstract method

Q208 **What is the correct syntax for varargs?**

a. **void method(int... nums) ✓**

b. void method(int nums...)

c. void method(int[] nums)

d. void method(*int nums)

Q209 **What is an enum in Java?**

a. A class with public fields

b. **A special data type representing a group of constants ✓**

c. An array of integers

d. A functional interface

Q210 **Can an enum have a constructor in Java?**

a. No

b. **Yes, but it must be private or package-private ✓**

c. Yes, and it can be public

d. Only abstract enums can

Q211 **What is the output of: enum Day {MON,TUE,WED;} System.out.println(Day.MON.ordinal());?**

a. 1

b. **0 ✓**

c. MON

d. Compile error

Q212 **What is the purpose of the Comparable interface's compareTo() method?**

a. Compares two strings only

b. **Returns negative, zero, or positive to indicate ordering ✓**

c. Returns a boolean

d. Sorts the collection

Q213 **Which method must be implemented when implementing the Comparator interface?**

a. compareTo()
b. equals()
c. compare() ✓
d. sort()

Q214 What is a stack in Java Collections?
a. FIFO data structure
b. LIFO data structure ✓
c. Sorted set
d. Double-ended queue

Q215 Which class implements a Queue in Java?
a. Stack
b. ArrayList
c. LinkedList ✓
d. HashSet

Q216 What does PriorityQueue use to order its elements?
a. Insertion order
b. Random order
c. Natural ordering or custom Comparator ✓
d. Alphabetical order only

Q217 What is the output of: Map<String,Integer> map=new HashMap<>(); map.put('a',1); map.put('a',2); System.out.println(map.get('a'));?
a. 1
b. 2 ✓
c. [1,2]
d. null

▶ LAMBDA & FUNCTIONAL INTERFACES (9 Questions)

Q218 Which functional interface takes input and produces output?
a. Consumer
b. Function ✓
c. Supplier
d. Predicate

Q219 Which of the following represents a valid lambda expression?

a. **() -> System.out.println('Hello') ✓**
b. -> System.out.println('Hello')
c. () System.out.println('Hello')
d. () => System.out.println('Hello')

Q220 **Predicate<Integer> p = x -> x > 10; System.out.println(p.test(15)); What will be the output?**

a. **true ✓**
b. false
c. 15
d. Compilation error

Q221 **Consumer<String> c = s -> System.out.println(s.toUpperCase()); c.accept('java'); Output?**

a. java
b. **JAVA ✓**
c. Java
d. Compilation error

Q222 **Which annotation is used to declare a functional interface?**

a. @Functional
b. @Lambda
c. **@FunctionalInterface ✓**
d. @Interface

Q223 **What is the return type of Predicate?**

a. int
b. **boolean ✓**
c. void
d. Object

Q224 **Which functional interface accepts input but returns nothing?**

a. **Consumer ✓**
b. Supplier
c. Predicate
d. Function

Q225 **Which functional interface does not take input but returns a value?**

a. Consumer
b. Predicate
c. **Supplier ✓**

d. Function

Q226 **Which package contains Predicate, Function, Consumer and Supplier?**

a. java.util
b. java.util.function ✓
c. java.util.stream
d. java.lang.function

▶ STREAMS API (18 Questions)

Q227 **Streams were introduced in which Java version?**

a. Java 6
b. Java 7
c. Java 8 ✓
d. Java 11

Q228 **Which package contains Stream API?**

a. java.util.stream ✓
b. java.stream
c. java.lang.stream
d. java.util.collection

Q229 **Which method converts a collection to a stream?**

a. toStream()
b. stream() ✓
c. convertStream()
d. getStream()

Q230 **Which stream operation filters elements?**

a. map
b. reduce
c. filter ✓
d. collect

Q231 **Which stream operation transforms elements?**

a. filter
b. map ✓
c. count
d. anyMatch

Q232 **Which stream operation combines elements to produce a single result?**

a. map
b. reduce ✓
c. filter
d. flatMap

Q233 **Which operation sorts stream elements?**

a. sorted() ✓
b. order()
c. arrange()
d. organize()

Q234 **Which stream method checks if any element matches a condition?**

a. match()
b. anyMatch() ✓
c. find()
d. test()

Q235 **Which operation is a terminal operation?**

a. filter
b. map
c. forEach ✓
d. sorted

Q236 **Which of the following is NOT a stream operation?**

a. filter
b. map
c. reduce
d. insert ✓

Q237 **Which operation flattens nested streams?**

a. map
b. flatMap ✓
c. reduce
d. sort

Q238 **Streams differ from collections because streams:**

a. Store data
b. Do not store data ✓

c. Are mutable
d. Are synchronized

Q239 **Which primitive stream works with int values?**

a. IntegerStream
b. IntStream ✓
c. IntCollection
d. StreamInt

Q240 **Which primitive stream works with long values?**

a. LongStream ✓
b. StreamLong
c. LongCollection
d. LStream

Q241 **Which primitive stream works with double values?**

a. DoubleCollection
b. DoubleStream ✓
c. StreamDouble
d. DStream

Q242 **Which method converts a primitive stream to a boxed stream?**

a. convert()
b. boxing()
c. boxed() ✓
d. wrap()

Q243 **IntStream.range(1,4).forEach(System.out::print); Output?**

a. 123 ✓
b. 1234
c. 0123
d. Compilation error

Q244 **IntStream.of(1,2,3).sum(); What is the result?**

a. 5
b. 6 ✓
c. 3
d. Compilation error

▶ GENERICS & COLLECTIONS FRAMEWORK (4 Questions)

Q245 **Which class is used for arbitrary precision integer arithmetic?**

a. Integer
b. BigInteger ✓
c. Math
d. Number

Q246 **Which feature allows defining type-safe collections?**

a. Serialization
b. Generics ✓
c. Lambda
d. Reflection

Q247 **Which package contains the Collections Framework?**

a. java.lang
b. java.util ✓
c. java.math
d. java.stream

Q248 **Which package is automatically imported in every Java program?**

a. java.util
b. java.lang ✓
c. java.math
d. java.io

▶ ADVANCED LAMBDA & STREAMS (40 Questions)

Q249 **Which of the following is a correct method reference to an instance method of an arbitrary object of a particular type?**

a. String::new
b. String::toUpperCase ✓
c. System.out::println
d. Integer::parseInt

Q250 **What is the output of: List.of(1,2,3,4).stream().filter(x->x%2==0).count();?**

a. 1
b. 2 ✓
c. 3
d. 4

Q251 **What does Stream.map() return?**

a. A boolean

b. A new Stream with transformed elements ✓

c. A List

d. void

Q252 **What is the key difference between map() and flatMap() in streams?**

a. No difference

b. map() transforms 1-to-1; flatMap() transforms 1-to-many and flattens the result ✓

c. flatMap() filters; map() flattens

d. map() is terminal; flatMap() is intermediate

Q253 **Which is the correct way to collect stream results into a List?**

a. .collect(Collectors.toList()) ✓

b. .toList(new ArrayList())

c. .asList()

d. .result()

Q254 **What does Collectors.groupingBy() do?**

a. Sorts elements

b. Groups stream elements by a classifier function into a Map ✓

c. Filters elements

d. Reduces elements to one value

Q255 **Which terminal operation returns an Optional containing the first element of the stream?**

a. findFirst() ✓

b. getFirst()

c. first()

d. peek()

Q256 **What does Stream.peek() do?**

a. Terminal operation that prints elements

b. Intermediate operation used for debugging without consuming the stream ✓

c. Same as forEach

d. Filters elements

Q257 **What is the output of: Stream.of(3,1,2).sorted().findFirst().get();?**

a. 3

b. 1 ✓

c. 2
d. Optional[3]

Q258 **Which method checks if ALL elements of a stream match a given predicate?**

a. anyMatch()
b. noneMatch()
c. allMatch() ✓
d. findAny()

Q259 **Which method checks if NO elements of a stream match a given predicate?**

a. allMatch()
b. noneMatch() ✓
c. anyMatch()
d. notMatch()

Q260 **What does Stream.distinct() do?**

a. Sorts the stream
b. Removes duplicate elements ✓
c. Filters odd elements
d. Counts elements

Q261 **What does Stream.limit(n) do?**

a. Skips first n elements
b. Returns a stream of at most n elements ✓
c. Filters elements greater than n
d. Throws exception if stream has fewer than n elements

Q262 **What does Stream.skip(n) do?**

a. Returns first n elements
b. Skips first n elements and returns the rest ✓
c. Removes last n elements
d. Limits to n elements

Q263 **What is the output of: Stream.of(1,2,3).reduce(0,(a,b)->a+b);?**

a. 0
b. Optional[6]
c. 6 ✓
d. Compilation error

Q264 **What does Optional.isPresent() return?**

a. The value inside Optional

b. true if value is present; false if empty ✓

c. void

d. A Stream

Q265 What does Optional.ifPresent(Consumer) do?

a. Returns the value

b. Throws exception if empty

c. Executes the Consumer only if a value is present ✓

d. Returns an empty Optional

Q266 What is the output of: Optional.empty().orElse('default');?

a. null

b. Optional.empty

c. default ✓

d. NoSuchElementException

Q267 Which of the following correctly chains two Predicates with AND logic?

a. p1.and(p2) ✓

b. p1 && p2

c. p1.plus(p2)

d. Predicate.and(p1,p2)

Q268 Which of the following correctly negates a Predicate p in Java?

a. !p

b. p.negate()

c. Predicate.not(p)

d. Both b and c ✓

Q269 BiFunction<Integer,Integer,Integer> add = (a,b)->a+b; System.out.println(add.apply(3,4)); Output?

a. 3

b. 4

c. 7 ✓

d. Compilation error

Q270 Which functional interface represents a function that accepts two arguments of possibly different types?

a. Function

b. BiFunction ✓

c. BiConsumer

d. BinaryOperator

Q271 **UnaryOperator<Integer> square = x -> x*x; System.out.println(square.apply(5)); Output?**

a. 5
b. 10
c. 25 ✓
d. Compilation error

Q272 **What does BinaryOperator<T> represent?**

a. A function taking one argument
b. A function taking two arguments of the SAME type and returning the same type ✓
c. A predicate for two values
d. A consumer for two values

Q273 **Supplier<String> s = () -> 'Hello'; System.out.println(s.get()); Output?**

a. null
b. Hello ✓
c. ()
d. Compilation error

Q274 **What is the output of: List.of('a','b','c').stream().map(String::toUpperCase).collect(Collectors.joining(',')); ?**

a. abc
b. A,B,C ✓
c. [A,B,C]
d. a,b,c

Q275 **What does Collectors.joining(delimiter) do?**

a. Adds elements to a Set
b. Concatenates stream String elements into one String using the delimiter ✓
c. Counts elements
d. Groups by length

Q276 **What does Collectors.counting() return as part of a collector?**

a. Integer
b. Long — the count of stream elements ✓
c. List
d. Map

Q277 **What is a parallel stream in Java?**

a. A stream that runs slower
b. A stream that processes elements concurrently using multiple threads ✓
c. A stream that filters in parallel only
d. A synchronized stream

Q278 How do you correctly create a parallel stream from a List in Java?
a. list.stream().parallel() or list.parallelStream() ✓
b. list.threadStream()
c. list.concurrentStream()
d. list.multiStream()

Q279 String::new in a method reference is an example of which type?
a. Static method reference
b. Instance method reference of a particular instance
c. Constructor reference ✓
d. Abstract method reference

Q280 What is the output of: IntStream.rangeClosed(1,5).sum();?
a. 10
b. 15 ✓
c. 14
d. 1

Q281 What is the difference between IntStream.range(1,5) and IntStream.rangeClosed(1,5)?
a. No difference
b. range excludes the end value (1,2,3,4); rangeClosed includes it (1,2,3,4,5) ✓
c. rangeClosed excludes the start
d. range includes both ends

Q282 Which Collector creates a Map where each stream element becomes a key mapped by a given function?
a. Collectors.toMap() ✓
b. Collectors.groupingBy()
c. Collectors.joining()
d. Collectors.toSet()

Q283 What does Stream.generate(Supplier) do?
a. Creates a finite stream from a Supplier
b. Creates an infinite sequential unordered stream using a Supplier ✓
c. Generates a random integer stream
d. Creates a stream from an array

Q284 What does Stream.iterate(0, x->x+2) do when used with limit()?

a. Creates a stream of exactly [0,2]

b. Creates a potentially infinite stream starting at 0 and incrementing by 2 each step ✓

c. Creates exactly 2 elements

d. Throws exception after 10 elements

Q285 Which is the correct way to convert a Stream<String> to a String[]?

a. .toArray(String[]::new) ✓

b. .collect(toArray())

c. .asArray()

d. .toStringArray()

Q286 What happens if you call a terminal operation on the same stream instance twice?

a. Returns same result both times

b. Throws IllegalStateException — stream has already been operated upon or closed ✓

c. Returns empty result

d. Creates a new stream automatically

Q287 What is 'lazy evaluation' in the context of Java Streams?

a. Streams always compute all results immediately

b. Intermediate operations are deferred and not executed until a terminal operation is invoked ✓

c. Streams skip null values lazily

d. Parallel streams are lazy by default

Q288 Which interface does Stream<T> directly extend in Java?

a. Iterable<T>

b. Collection<T>

c. BaseStream<T,Stream<T>> ✓

d. Closeable

✅ **Total: 288 Questions Completed**

Green highlighted option = Correct Answer